卷册检索号 SEARCH NO. OF THE VOLUME
MLHVDC-LCS-ECI-DD-CM-0302 30-B003701S-U0302

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图纸文件目录（首页）

CATALOG OF DRAWING DOCUMENTS (HEAD PAGE)

拉合尔±660kV换流站 工程 施工图 阶段
±660kV Lahore Converter Station PROJECT Detail Design DESIGN STAGE

通信 专业 第 03 卷 第 02 册
Communication SPECIALITY VOLUME BOOK

卷册名称 保护接口柜与通信设备安装图
Volume name Detail Design Drawings of Protection Interface Equipment with Communication Equipment

批 准 APPROVED BY	
审 核 REVIEWED BY	
设计/勘测 DESIGNED BY/ SURVEYED BY	
图纸 9 张 DRAWING NOS.	打印件 0 本 PRINTED COPIES NOS.

序号 No.	图 号 DRAWING NO.	文件名称 DOCUMENT TITLE	版本 EDITION	张 NOS.	本 NOS.	备注 REMARKS
1	30-B003701S-U0302-001	设计说明 Detail Design Descriptions	A	1		
2	30-B003701S-U0302-002	设备材料清单 List of Equipments and Materials	A	1		
3	30-B003701S-U0302-003	通信机房平面布置图 Layout Diagram of Communication Room	A	1		
4	30-B003701S-U0302-004	交流侧保护接口方式系统连接图 Teleprotection interconnection for trip scheme for AC outgoing lines	A	1		
5	30-B003701S-U0302-005	拉合尔换流站系统通信方案图 Communication System Scheme of Lahore Converter Station	A	1		
6	30-B003701S-U0302-006	拉合尔换流站500kV交流系统通信方案图 500kV AC Communication System Scheme of Lahore Converter Station	A	1		
7	30-B003701S-U0302-007	保护接口屏柜布置图 Teleprotection Interface Equipment Panel Drawing	A	1		
8	30-B003701S-U0302-008	保护接口设备面板图 Teleprotection Interface Equipment Panel Drawing	A	1		
9	30-B003701S-U0302-009	TPS-NU端子图 TPS-NU Terminals Drawing	A	1		

注:

1. 拉合尔换流站本体通信的设计范围主要包括：换流站通信电源系统、站内交流电力线载波系统、调度及行政程控交换系统及站内综合布线。
1. The design scope of communication of Lahore converter station in this project mainly includes: communication equipment, AC power line carrier system, communication power supply system, communication room, PABX switching system and Information Network of cabling system.

2. 本工程拉合尔换流站通信施工图共3卷6册，分别为：
2. There are 6 volumes detailed drawings of Lahore converter station as follows:

30-B003701S-U0101：通信-48V直流电源及蓄电池系统安装图；
Detail Design Drawings of -48V DC Power Equipment and Battery for Communication of Lahore

30-B003701S-U0102：调度及行政交换机系统安装图；
Detail Design Drawings of Dispatch and Administration Switch

30-B003701S-U0201：站内信息网络设备安装图；
Detail Design Drawings of Informatica Network Equipment of Lahore

30-B003701S-U0202：站内综合布线安装图；
Detail Design Drawings of Cabling System of Lahore

30-B003701S-U0301：交流电力线载波机设备安装图；
Detail Design Drawings of PLC Equipment for AC Power Line

30-B003701S-U0302：保护接口柜与通信设备安装图。
Detail Design Drawings of Protection Interface Equipment with Communication Equipment

本卷册为：30-B003701S-U0302：保护接口柜与通信设备安装图，主要为拉合尔换流站站内保护接口屏的安装和连接工作。
This volume is Mainly for the installation and connection of Teleprotection equipment in the Lahore Converter Station.

3. 本工程拉合尔换流站交流出线共3回，2回出线至500kV拉合尔变，2回出线至500kV拉合尔南变，2回远景出线规划至500kV拉合尔北变。根据NTDC控保部门的反馈，拉合尔换流站到500kV拉合尔南变的保护采用光纤专用芯的方式，到500kV拉合尔变和500kV拉合尔北变采用2套PLC+1套SDH传输通道的方式。保护接口装置屏请见卷册内平面布置图。
3. There are three AC outgoing lines for Lahore Converter Station, 2 circuits to 500kV Lahore Station, 2 circuits to 500kV Lahore South Station and 2 planning circuits to 500kV Lahore North Station. According to NTDC Protection Department, to 500kV Lahore South shall use optical fiber for teleprotection, and to 500kV Lahore Station and Lahore North station will use 2 sets of PLC and 1 set of SDH for each circuit.

4. 本工程500kV拉合尔南变出线方向的保护方式，经NTDC控保及Telcom部门要求本期使用专用芯方式传输保护信号。具体光纤纤芯分配及连接设备请见光通信施工图。
4. Teleprotcetion of 500kV Lahore South Station will use Optical fiber with 1 main 1 backup by NTDC Protection Department. Details optical fibers connection please see Optical Communication Drawings.

5. 本工程所选设备、材料，必须具有国家级检测中心的检测合格证书（3C认证）；必须满足与产品相关的国家标准；供电产品、消防产品应具有入网许可证。
5. The equipment and materials selected for this project must have the qualification certificate of national testing center (3C certification).and must meet national standards relating to products;Power supply products and fire products shall have access to the Internet license.

6. 所选设备型号仅供参考，招标所确定的设备规格、性能等技术指标，不应低于设计图纸的要求。
6. The selected equipment model is for reference only,The specifications, performance and other technical indicators specified in the bidding shall not be lower than the requirements of the design drawings.

7. 本卷册设计以中文部分为主，英文部分仅作参考。
7. This volume is mainly designed in Chinese, and the English part is for reference only.

巴基斯坦默蒂亚里—拉合尔±660kV直流输电工程 ±660kV HVDC Project from Matiari to Lahore in Pakistan		NATIONAL TRANSMISSION & DESPATCH COMPANY (NTDC) LIMITED PAKISTAN	
CONTRACTOR 国家电网 STATE GRID 中国电力技术装备有限公司 CHINA ELECTRIC POWER EQUIPMENT AND TECHNOLOGY CO.,LTD.		SUBCONTRACTOR East China Electric Power Design Institute Co.,Ltd.	
拉合尔±660kV换流站 ±660kV Lahore Converter Station		设计阶段 PHASE	施工图 Detail Design
M L H V D C - L C S - E C I - D D - C M - 0 3 0 2 - 0 0 1 A			
	姓名 Name	日期 Date	设计说明 Detail Design Descriptions
设计 Designed		2020.02.15	
校核 Checked		2020.02.15	
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比例 Scale	1:1	图号 DWG NO.	30-B003701S-U0302-001
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420X594

序号 No.	设备名称 Name of Equipment	型号及规范 Specification and Type	单位 Unit	数量 Qty	供货 Supply	备注 Notes
1	SELTA 保护接口装置 SELTA teleprotection equipment					
1.1	保护接口装置 teleprotection board	TPS-NU	套 Set	4	黑马 HIMARK	
1.2	机柜 Panel	600*600*2260	套 Set	1	黑马 HIMARK	

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拉合尔±660kV换流站

±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detail Design

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设备材料清单

List of Equipments and Materials

比例 Scale	1:1	图号 DWG NO.	30-B003701S-U0302-002	版本 REV.	A

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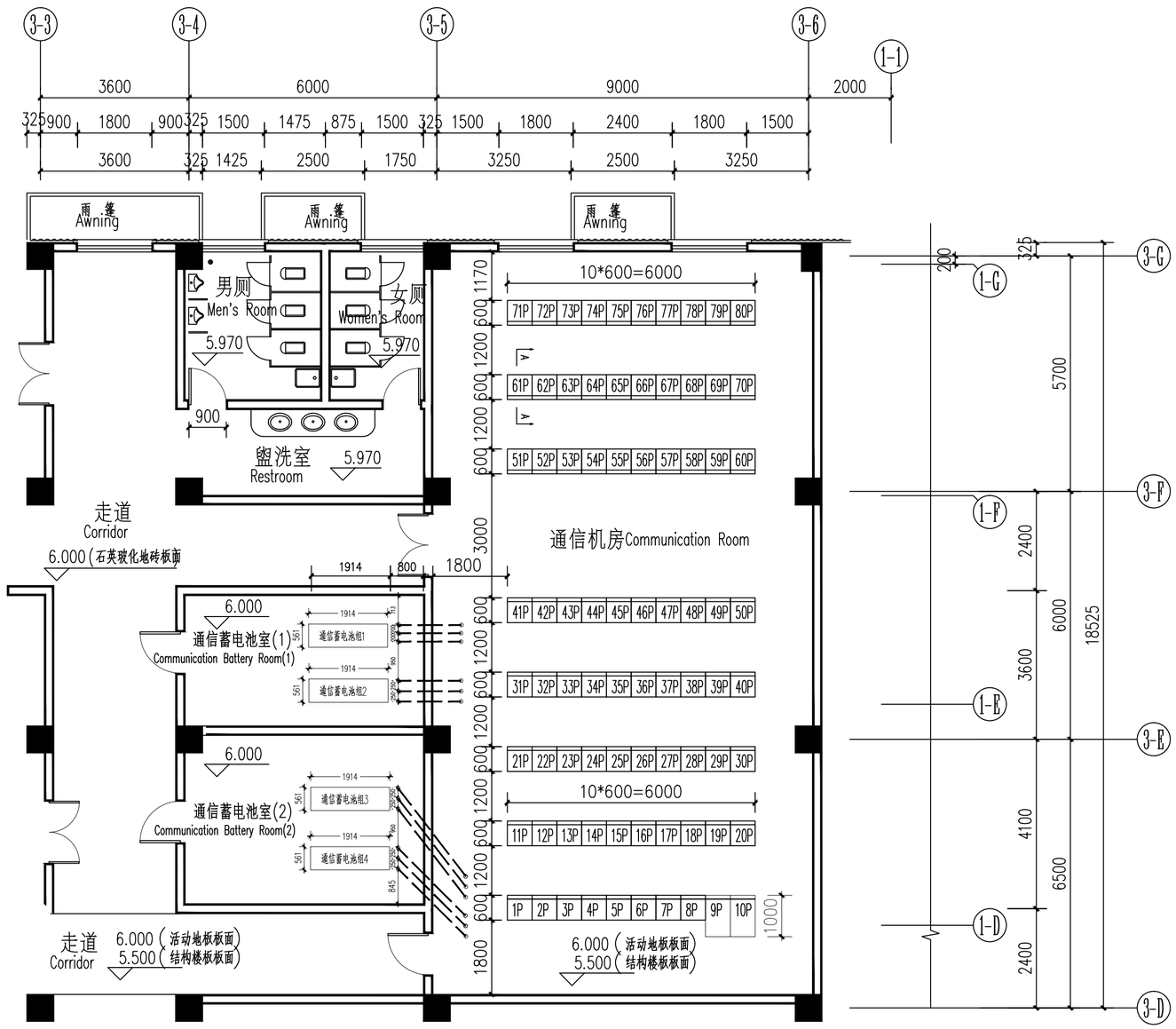

控制楼6.000m层平面图/Control Building 6.000m Plan

序号 NO.	名称 NAME	型号 SPECIFICATION	数量 QT'Y
1P	光纤配线屏1 ODF PANEL 1	ODF-24*7	1
2P	光纤配线屏2 ODF PANEL 2	ODF-24*7	1
3P	光路子系统屏1 OPTICAL AMPLIFIER PANEL 1	Optix OSN 9800	1
4P	光路子系统屏2 OPTICAL AMPLIFIER PANEL 2	Optix OSN 9800	1
5P	SDH(内置PCM)屏1 SDH (INTERNAL PCM) PANEL 1	OptiX OSN 1800V	1
6P	SDH(内置PCM)屏2 SDH (INTERNAL PCM) PANEL 2	OptiX OSN 1800V	1
7P	数字配线屏 DDF PANEL	DDF-16*8	1
8P	音频配线屏3 VDF PANEL 3	VDF-200	1
9P	SDH网管服务器屏 SDH NETWORK MANAGEMENT SERVER PANEL	600*1000*2260	1
10P	动力环境监控服务器屏 POWER ENVIRONMENT MONITOR PANEL	600*1000*2260	1
11P	行政交换机屏 ADMINISTRATION PABX PANEL	ALE OminiPCX	1
12P	录音系统屏 RECORD SYSTEM PANEL	600*600*2260	1
13P	音频配线屏1 VDF PANEL 1	VDF-100*5	1
14P	音频配线屏2 VDF PANEL 2	VDF-100*5	1
15P	网络设备屏1 NETWORK EQUIPMENT PANEL 1	600*600*2260	1
16P	网络设备屏2 NETWORK EQUIPMENT PANEL 2	600*600*2260	1
17P	线路保护接口屏1 TELEPROTECTION EQUIPMENT PANEL 1	SELTA TPS-NU	1
18P	调度交换机屏 DISPATCHING PABX PANEL	SELTA SAM 4000E	1
19P-20P	备用屏位 Spare Panel	600*600*2260	2
21P	通信电源柜1 High Frequency Switching Power Supply 1	ZMT48V/500A	1
22P	直流配电柜1 DC distribution panel 1	HIMARK	1
23P	直流配电柜2 DC distribution panel 2	HIMARK	1
24P	通信电源柜2 High Frequency Switching Power Supply 2	ZMT48V/500A	1
25P	电能计费通信接口屏 COMMUNICATION INTERFACE PANEL FOR METERING SYSTEM	600*600*2260	1
26P-30P	备用屏位 Spare Panel	600*600*2260	5
31P	电力线载波机屏1-拉合尔变1回 PLC PANEL 1-LAHORE LINE 1	SELTA STE-D + TPS-NU	1
32P	电力线载波机屏2-拉合尔变1回 PLC PANEL 2-LAHORE LINE 1	SELTA STE-D + TPS-NU	1
33P	电力线载波机屏1-拉合尔变2回 PLC PANEL 1-LAHORE LINE 2	SELTA STE-D + TPS-NU	1
34P	电力线载波机屏2-拉合尔变2回 PLC PANEL 2-LAHORE LINE 2	SELTA STE-D + TPS-NU	1
35P	电力线载波机屏1-拉合尔南变1回 PLC PANEL 1-LAHORE SOUTH LINE 1	SELTA STE-D + TPS-NU	1
36P	电力线载波机屏2-拉合尔南变1回 PLC PANEL 2-LAHORE SOUTH LINE 1	SELTA STE-D + TPS-NU	1
37P	电力线载波机屏1-拉合尔南变2回 PLC PANEL 1-LAHORE SOUTH LINE 2	SELTA STE-D + TPS-NU	1
38P	电力线载波机屏2-拉合尔南变2回 PLC PANEL 2-LAHORE SOUTH LINE 2	SELTA STE-D + TPS-NU	1
39P	备用屏位 Spare Panel	600*600*2260	1
40P	备用屏位 Spare Panel	600*600*2260	1
41P	电力线载波机屏1-拉合尔北变1回 PLC PANEL 1-LAHORE NORTH LINE 1	SIEMENS POWERLINK100	1
42P	电力线载波机屏2-拉合尔北变1回 PLC PANEL 2-LAHORE NORTH LINE 1	SIEMENS POWERLINK100	1
43P	电力线载波机屏1-拉合尔北变2回 PLC PANEL 1-LAHORE NORTH LINE 2	SIEMENS POWERLINK100	1
44P	电力线载波机屏2-拉合尔北变2回 PLC PANEL 2-LAHORE NORTH LINE 2	SIEMENS POWERLINK100	1
45P-50P	备用屏位 Spare Panel	600*600*2260	6
51P-80P	备用屏位 Spare Panel	600*600*2260	30
	通信蓄电池室(1) Communication Battery Room(1)		
1	通信蓄电池组1 Communication Battery 1	GFMH600/48V	1
2	通信蓄电池组2 Communication Battery 2	GFMH600/48V	1
	通信蓄电池室(2) Communication Battery Room(2)		
1	通信蓄电池组3 Communication Battery 3	GFMH600/48V	1
2	通信蓄电池组4 Communication Battery 4	GFMH600/48V	1

巴基斯坦默蒂亚里—拉合尔±660kV直流输电工程
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拉合尔±660kV换流站
±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detail Design

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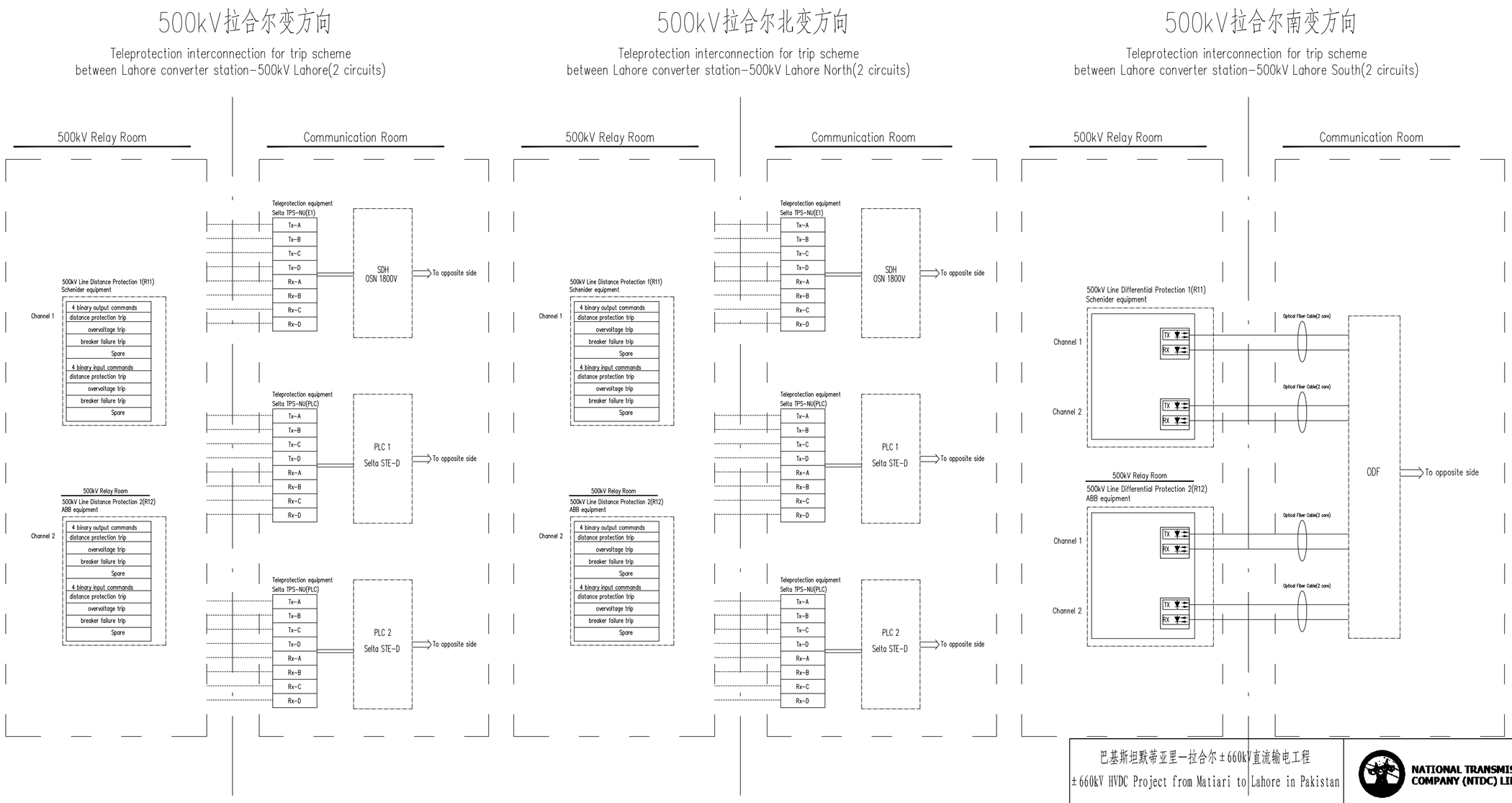
通信机房平面布置图
Layout Diagram of Communication Room

比例 Scale	1:1	图号 DWG NO.	30-B003701S-U0302-003	版本 REV.	A

420X594



Note: 4 Line Differential Relays (for 2 lines between Lahore converter station and 500kV Lahore South substation) will use 8 fiber cores of the OPGW (1 main and 1 backup).

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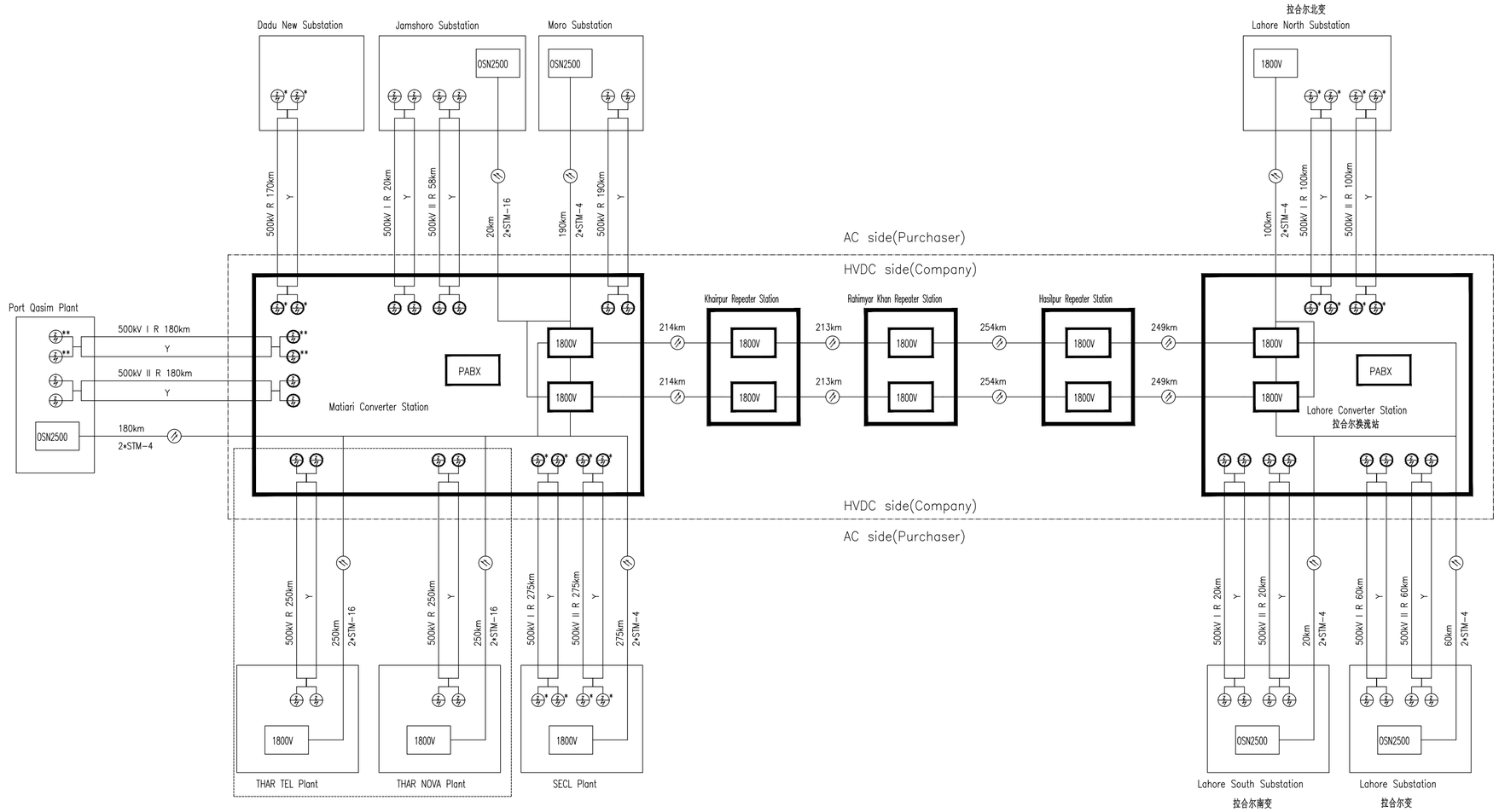
交流侧保护接口方式系统连接图

Teleprotection interconnection for trip scheme for AC outgoing lines

比例 Scale	1:1	图号 DWG NO.	30-B003701S-U0302-004	版本 REV.	A

420X594



Legend:

- PLC terminal(Selta STE-D)
- PLC terminal(Siemens Powerlink100)
- PLC terminal(ABB ETL81)
- SDH terminal(Huawei 1800V)
- SDH terminal(Huawei OSN2500)
- PABX exchange(Alcatel)
- OPGW

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±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detail Design

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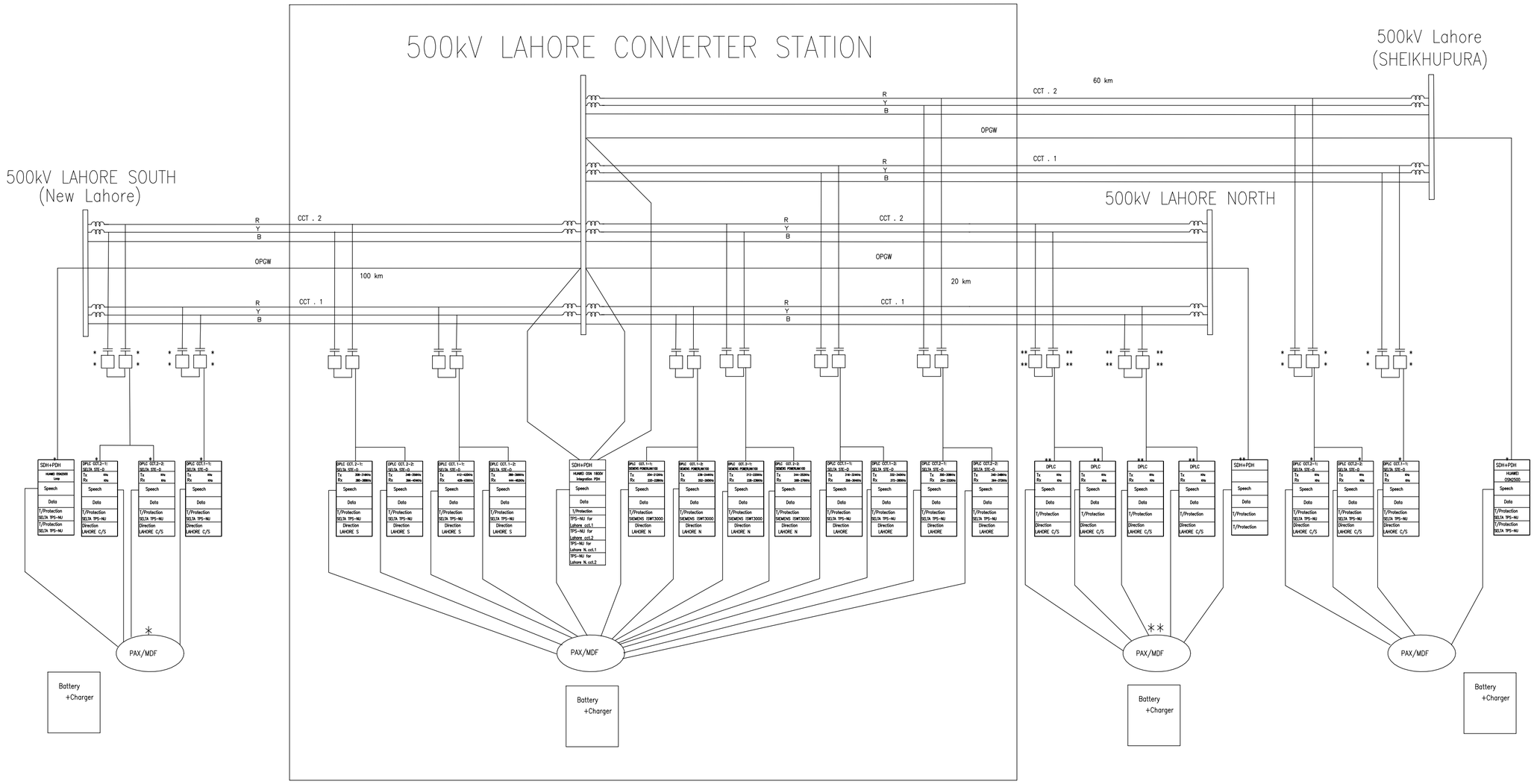
拉合尔换流站系统通信方案图

Communication System Scheme of Lahore Converter Station

比例 Scale	1:1	图号 DWG NO.	30-B003701S-U0302-005	版本 REV.	A

420X594



* EXISTING EQUIPMENT

** Planning station, should compatible with Lahore C/S.

Note: CET is responsible for the equipment inside the Lahore Convert Station. Outside is responsible for NTDC.

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拉合尔换流站500kV交流系统通信方案图

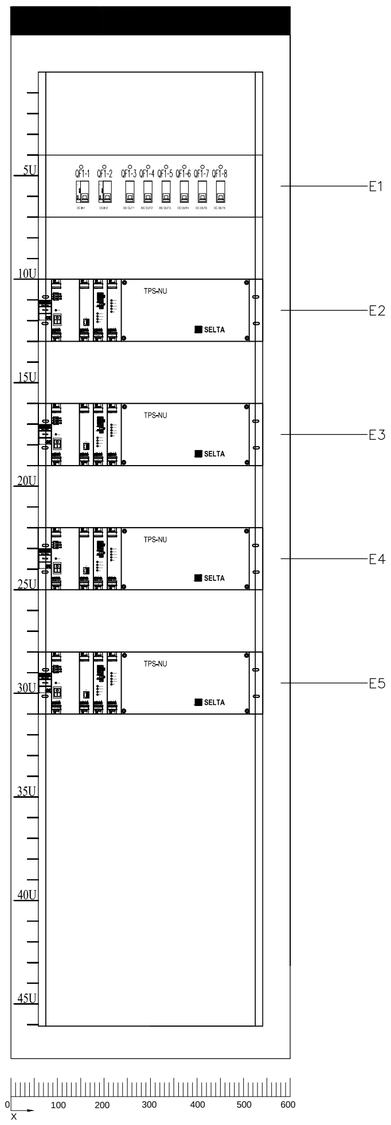
500kV AC Communication System Scheme of Lahore Converter Station

比例 Scale	1:1	图号 DWG NO.	30-B003701S-U0302-006	版本 REV.	A

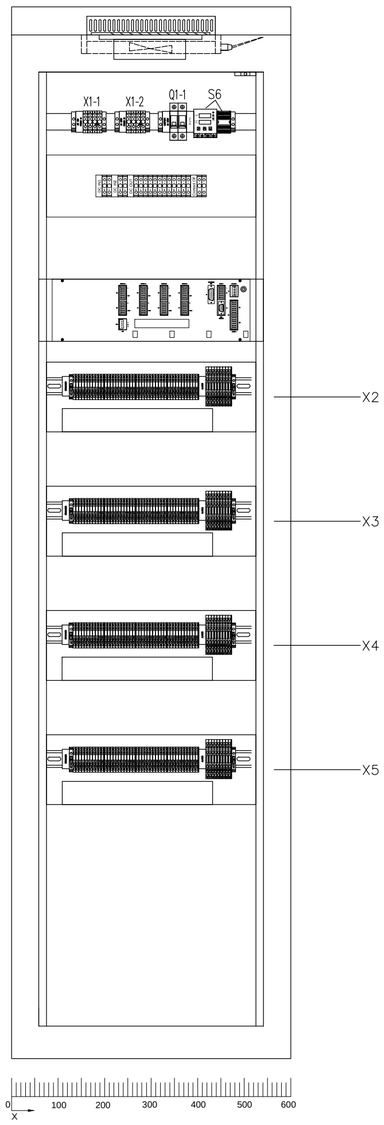
420X594

序号 Item. No.	名称 Name	规格 Type
E1	电源分配单元 Power Distribution Unit	PDU-6
E2~E5	保护接口设备 Teleprotection Interface Equipment	Selta TPS-NU
X1-1,X1-2	电源进线端子 Power Supply Incoming Terminals	10N
X2~X5	TPS-NU端子 TPS-NU Terminals	
Q1-1	微型断路器 Miniature Circuit Breaker	MCB:CHINT AC 230V,6A/2P
QF1-1,QF1-2	微型断路器 Miniature Circuit Breaker	MCB:CHINT DC 48V,10A/1P
QF1-3~QF1-8	微型断路器 Miniature Circuit Breaker	MCB:CHINT DC 48V,6A/1P
S6	温湿度控制器 Temperature & Humidity Controller	WSK1-1DS-J
S7	加热器 Heater	DJR,CHINA

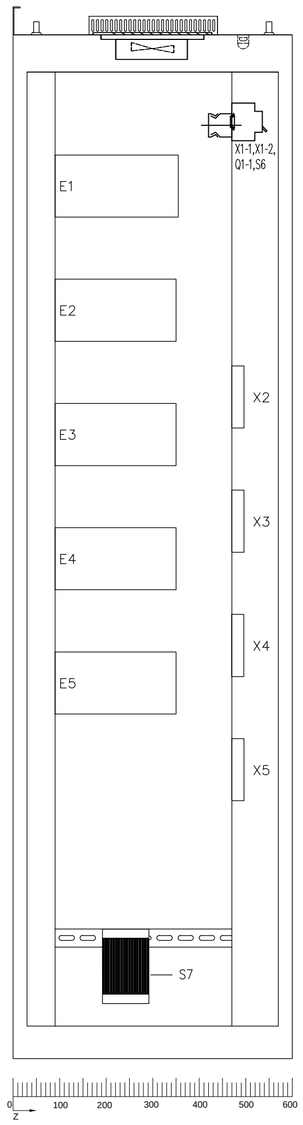
17P 线路保护接口屏1 TELEPROTECTION EQUIPMENT PANEL 1



前视图
Front View



后视图
Rear View



侧视图
Side View

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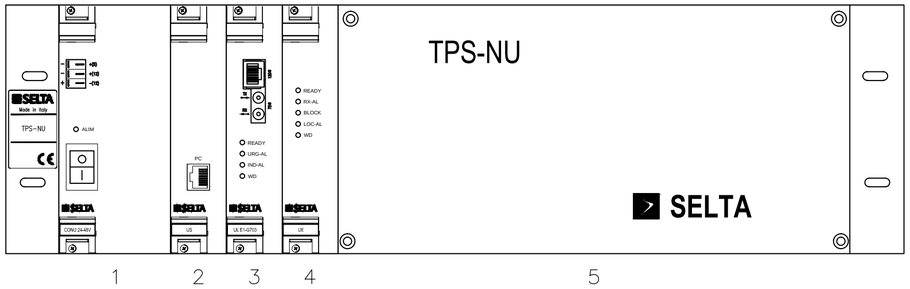
保护接口屏柜布置图
Teleprotection Interface Equipment Panel Drawing

比例 Scale: 1:1
图号 DWG NO.: 30-B003701S-U0302-007
版本 REV.: A

420X594

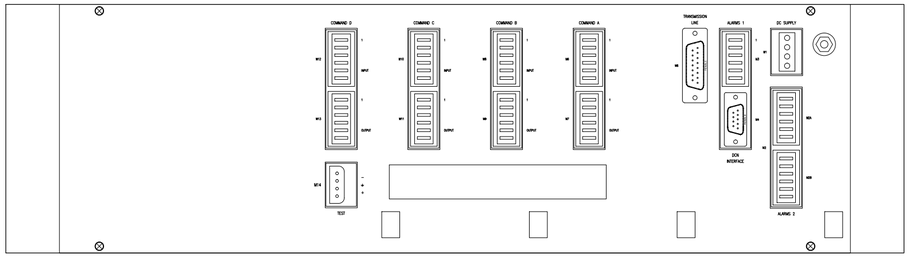
SELTA TPS-NU前视图
SELTA TPS-NU Front View



设备面板配置表
Equipment Panel Configuration Table

序号 No.				
		CONU 24-48V		

SELTA TPS-NU后视图
SELTA TPS-NU Rear View



巴基斯坦默蒂亚里—拉合尔±660kV直流输电工程
±660kV HVDC Project from Matiari to Lahore in Pakistan

NATIONAL TRANSMISSION & DESPATCH COMPANY (NTDC) LIMITED PAKISTAN

CONTRACTOR 国家电网 STATE GRID 中国电力技术装备有限公司 CHINA ELECTRIC POWER EQUIPMENT AND TECHNOLOGY CO.,LTD.

SUBCONTRACTOR East China Electric Power Design Institute Co.,Ltd.

拉合尔±660kV换流站
±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detail Design

M L H V D C - L C S - E C I - D D - C M - 0 3 0 2 - 0 0 8 A

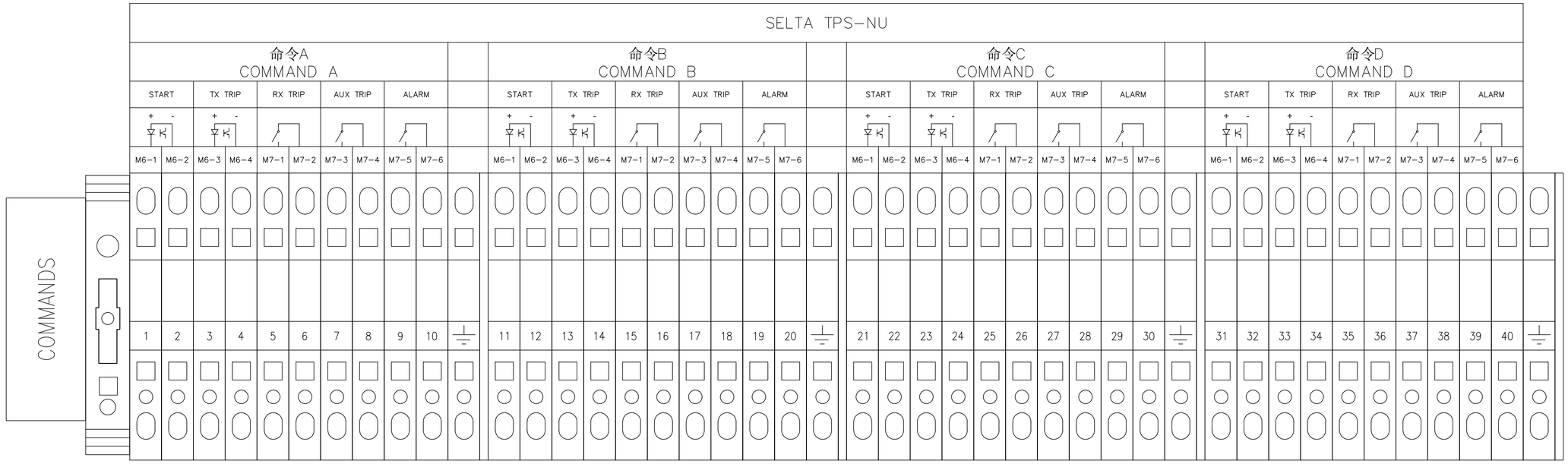
	姓名 Name	日期 Date
设计 Designed		2020.02.15
校核 Checked		2020.02.15
审核 Reviewed		2020.02.15
批准 Approved		2020.02.15

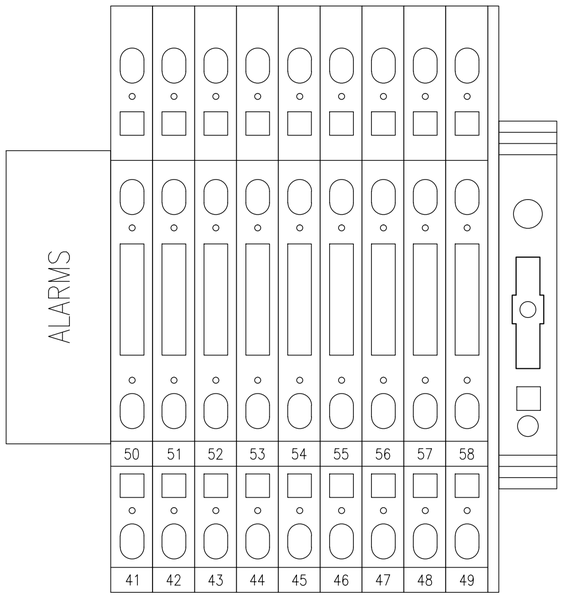
保护接口设备面板图
Teleprotection Interface Equipment Panel Drawing

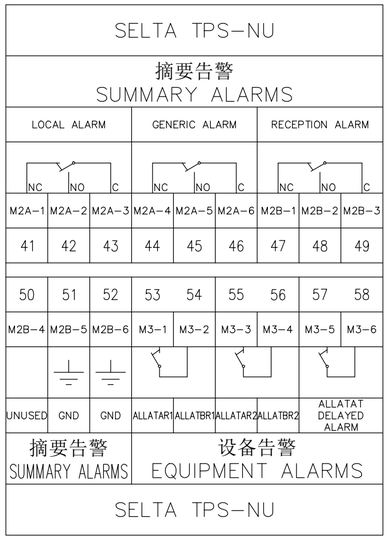
比例 Scale	1:1	图号 DWG NO.	30-B003701S-U0302-008	版本 REV.	A

420X594

X2/X3/X4/X5







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拉合尔±660kV换流站

±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detail Design

M L H V D C - L C S - E C I - D D - C M - 0 3 0 2 - 0 0 9 A

	姓名 Name	日期 Date
设计 Designed		2020.02.15
校核 Checked		2020.02.15
审核 Reviewed		2020.02.15
批准 Approved		2020.02.15

TPS-NU端子图

TPS-NU Terminals Drawing

比例 Scale	1:1	图号 DWG NO.	30-B003701S-U0302-009	版本 REV.	A

420X594